

Extending the May-15 inductive reduction to $r = 1$ shapes: partial generalisation and the same-row obstruction

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Abstract

The May-15 paper (2026-05-15-general-a-inductive.tex) proves a clean inductive reduction

$$B_{\ell+1}^{(\lambda)} V_\lambda = (T_{\ell+1}+1)(T_{\ell+1}+1) \cdots (T_{n-2}+1) \Phi_* \left(B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1} \right)$$

for the specific family $\lambda = (2^a, 1^{n-2a})$, with $\mu_1 = (2^{a-1}, 1^{n-2a})$. One might hope to generalise the May-15 argument verbatim to every rank-zero λ with $r = 1$ length-preserving corner. *This fails as stated.* The May-15 proof uses the equality $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda} = \Phi_*(V_{\mu_1})$, which holds for $(2^a, 1^{n-2a})$ but *not* for $r = 1$ shapes with same-row corner pairs.

Concrete counter-example to the naive generalisation. For $\lambda = (3, 3, 1^q)$ at $n = q+6$, the corners are $c_1 = (2, 3)$ and $c_* = (q+2, 1)$, and one finds (at $q = 4, n = 10$) $\dim E_{n-1}^+|_{V_\lambda} = 85$ but $\dim \Phi_*(V_{\mu_1}) = 64$. The 21-dimensional gap is filled by a *same-row part*: seminormal vectors at SYTs with letters $n-1, n$ at the same-row pair $(2, 2), (2, 3)$ of λ (a valid same-row configuration when $\lambda_2 = 3$ and $\lambda_3 = 1 < 2$).

What we prove. We isolate the correct general statement at $r = 1$:

- (1) The reduction formula at the level of $\Phi_*(V_{\mu_1})$ goes through verbatim from May-15 (Lemma 6):

$$R'_{\ell+1} R'_{\ell+2} \cdots R'_n|_{\Phi_*(V_{\mu_1})} = (T_{\ell+1}+1) \cdots (T_{n-2}+1) \Phi_*(B^{(\mu_1)}).$$

- (2) The May-19 Phase A endpoint gives $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$ unconditionally. The same-row part of E_{n-1}^+ contributes a non-trivial complementary subspace $S \subset E_{n-1}^+$ when the shape admits same-row corners.
- (3) A *full* reduction formula requires the additional claim $R'_{\ell+1} \cdots R'_{n-1} S = 0$ (the same-row part dies under the rest of the iteration). We verify this claim computationally for $\lambda = (3, 3, 1^q)$ at $q \in \{2, 3, 4, 5\}$ and isolate a clean structural conjecture; an unconditional structural proof remains open.

Application (partial). For $\lambda = (3, 3, 1^q)$ at $q \in \{2, 3, 4, 5\}$,

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = 2,$$

realising the May-13 closed-form prediction $\dim B = f^{(2,2)} = 2$. The proof combines the May-15-style commutation on the Φ_* -piece (rigorous), May-12 evening for the $\dim$ of $B^{(\mu_1)}$ (rigorous at $q \geq 4$, computational at $q = 2, 3$), and the same-row-dies claim (computational at $q \in \{2, 3, 4, 5\}$).

This paper is more an *anatomical study* of why the May-15 machinery does and does not generalise than a fresh structural theorem. The honest content is the isolation of the same-row obstruction and the discovery that it nevertheless vanishes in the relevant iterated image — a striking computational fact whose structural proof we have not located.

1 Setup and notation

Notation follows the May-13 multi-corner paper. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$. For $\lambda \vdash n$, V_λ is the seminormal Specht module with basis $\{v_T : T \in \text{SYT}(\lambda)\}$. Write $E_j^\pm := E_j^\pm(V_\lambda)$ for the $\pm$ -eigenspaces of T_j on V_λ . Define

$$R'_k := (T_{k-1}+1)(T_{k-2}+1)\cdots(T_1+1), \quad B_j^{(\lambda)}V_\lambda := R'_j R'_{j+1} \cdots R'_n V_\lambda.$$

The sharp index for λ is $j_0(\lambda) := \ell(\lambda) + 1$.

1.1 Hypotheses

Throughout the paper, fix $\lambda \vdash n$ satisfying:

- (H1) **One length-preserving corner.** λ has exactly $r = 1$ length-preserving corner $c_1 = (\rho, \lambda_\rho)$ with $1 \leq \rho < \ell(\lambda)$ and $\lambda_{\rho+1} < \lambda_\rho$.
- (H2) **Column-1 bottom.** $\lambda_\ell = 1$, so the column-1 bottom $c_* := (\ell, 1)$ is a removable corner.
- (H3) **Non-degenerate $\{c_1, c_*\}$ 2-block.** The axial distance $d := c(c_*) - c(c_1) = (1 - \ell) - (\lambda_\rho - \rho)$ satisfies $|d| \geq 2$.

These guarantee λ has exactly $r + 1 = 2$ removable corners. Let

$$\mu_1 := \lambda \setminus \{c_1, c_*\} \vdash n - 2,$$

so $\ell(\mu_1) = \ell - 1$.

2 The branching iso Φ_* and the E_+ gap

2.1 Φ_* on the corner-pair 2-blocks

Definition 1 (Φ_* at a corner pair). For each SYT $T' \in \text{SYT}(\mu_1)$ on letters $\{1, \dots, n-2\}$, define two SYTs of λ :

- $T_A := T' \cup \{n-1 \mapsto c_1, n \mapsto c_*\}$;
- $T_B := T' \cup \{n \mapsto c_1, n-1 \mapsto c_*\}$.

The pair $\{v_{T_A}, v_{T_B}\}$ spans a T_{n-1} -2-block of V_λ at axial distance d . Define $\Phi_*(v_{T'})$ as the $+q$ -eigenvector of this 2-block, with the coefficient of v_{T_A} normalised to 1; extend Φ_* linearly to V_{μ_1} .

Lemma 2 (Properties of Φ_*). $\Phi_* : V_{\mu_1}|_{n-2} \hookrightarrow E_{n-1}^+|_{V_\lambda}$ is a $\mathbb{Q}(q)$ -linear injection, $H_q(S_{n-2})$ -equivariant under $\langle T_1, \dots, T_{n-3} \rangle$.

Proof. Verbatim May-15 Lemmas 3.2–3.4 (2026-05-15-general-a-inductive.tex): T_i for $i \leq n-3$ commutes with T_{n-1} (index gap ≥ 2), and the positions of letters $1, \dots, n-2$ in T_A and T_B coincide, so T_i acts the same on both. Hence $T_i \Phi_*(v_{T'}) = \Phi_*(T_i v_{T'})$. $\square$

2.2 The E_+ gap: same-row pairs

Crucially, Φ_* may be a *strict* injection $\Phi_*(V_{\mu_1}) \subsetneq E_{n-1}^+|_{V_\lambda}$.

Lemma 3 (Decomposition of E_+ at $r = 1$). *At $r = 1$,*

$$E_{n-1}^+|_{V_\lambda} = \Phi_*(V_{\mu_1}) \oplus S, \quad (1)$$

where S is the span of seminormal basis vectors v_T for SYTs T of λ with the letters $\{n-1, n\}$ filling a same-row pair $\{(i, \lambda_i - 1), (i, \lambda_i)\}$ for some $1 \leq i \leq \ell$. The same-row pair is valid (and contributes to E_+) precisely when $\lambda_i \geq 2$ and either $i = \ell$ or $\lambda_{i+1} \leq \lambda_i - 2$.

Proof. T_{n-1} on a SYT T acts by the Hoefsmit rule, depending on the positions of letters $n-1, n$:

- Same row: $T_{n-1}v_T = qv_T$ (a $+q$ -eigenvector).
- Same column: $T_{n-1}v_T = -v_T$ (a -1 -eigenvector).
- Different row and column: a T_{n-1} -2-block at axial distance $d_T \in \mathbb{Z}$, with eigenvalues $\{q, -1\}$, giving one $+q$ -eigenvector.

Hence $E_{n-1}^+|_{V_\lambda}$ has a basis consisting of: (i) the $+q$ -eigenvectors from all T_{n-1} -2-blocks (one per unordered pair of corners), and (ii) all SYTs with $\{n-1, n\}$ in a valid same-row configuration.

For $r = 1$, the unordered pairs of corners reduce to the single pair $\{c_1, c_\ast\}$, and these $+q$ -eigenvectors are exactly the $\Phi_*(v_{T'})$ for $T' \in \text{SYT}(\mu_1)$.

The validity of a same-row pair at row i : requires $\lambda_i \geq 2$ (so the pair $(i, \lambda_i - 1), (i, \lambda_i)$ exists) and that no constraint forces a letter $> n$ in any cell. The latter requires that the cells of column $\lambda_i - 1$ below row i (i.e., $(i+1, \lambda_i - 1), \dots$) be unrestricted — concretely, $\lambda_{i+1} \leq \lambda_i - 2$ (so $(i+1, \lambda_i - 1)$ is not in λ), or $i = \ell$. $\square$

2.3 The $(3, 3, 1^q)$ same-row part

Example 4 ($(3, 3, 1^q)$ has a non-trivial same-row part). For $\lambda = (3, 3, 1^q)$ at $n = q + 6$: $\lambda_2 = 3$, $\lambda_3 = 1$, so $\lambda_3 \leq \lambda_2 - 2$. The same-row pair at row 2 is valid: SYTs with $n-1$ at $(2, 2)$ and n at $(2, 3)$. The remaining cells $\{(1, 1), (1, 2), (1, 3), (2, 1), (3, 1), \dots, (q+2, 1)\}$ form the sub-partition shape $(3, 1^{q+1})$ on $n-2 = q+4$ letters. Hence

$$\dim S = f^{(3, 1^{q+1})}.$$

For $q = 4$: $\dim S = f^{(3, 1^5)} = 21$. The full $E_+|_{V_\lambda}$ has $\dim \dim V_{\mu_1} + \dim S = 64 + 21 = 85$.

Remark 5 ($(2^a, 1^*)$ has trivial same-row part). For $\lambda = (2^a, 1^{n-2a})$: $\lambda_i = 2$ for $i \leq a$ and $\lambda_i = 1$ for $i > a$. Same-row pair at row $i \leq a$ requires $\lambda_{i+1} \leq 0$, i.e., $i = \ell$. But $\lambda_\ell = 1 \neq 2$ (when $\ell > a$), so no row i admits a same-row pair. Hence $S = 0$ for the $(2^a, 1^*)$ family, and the May-15 reduction goes through verbatim with $E_+ = \Phi_*(V_{\mu_1})$.

3 The May-15-style commutation on $\Phi_*(V_{\mu_1})$

The May-15 commutation argument is purely algebraic (index-gap based) and goes through on the Φ_* -piece of E_+ at every $r = 1$ shape.

Lemma 6 (Commutation through Φ_*). *At every $r = 1$ shape satisfying (H1)–(H3),*

$$R'_{\ell+1} R'_{\ell+2} \cdots R'_{n-1} |_{\Phi_*(V_{\mu_1})} = (T_\ell + 1)(T_{\ell+1} + 1) \cdots (T_{n-2} + 1) \Phi_* \left(R'_\ell^{(\mu_1)} R'_{\ell+1}^{(\mu_1)} \cdots R'_{n-2}^{(\mu_1)} \right). \quad (2)$$

In particular,

$$R'_{\ell+1} \cdots R'_{n-1} \Phi_*(V_{\mu_1}) = (T_\ell + 1) \cdots (T_{n-2} + 1) \Phi_*(B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1}).$$

Proof. The proof is verbatim the May-15 inductive commutation (2026-05-15-general-a-inductive.tex, Theorem 3.5), restricted to the Φ_* -image inside V_λ . The argument uses only:

- Φ_* is $H_q(S_{n-2})$ -equivariant (Lemma 2);
- the index-gap identity $R'_k = (T_{k-1}+1)R'_{k-1}$ together with the fact that R'_{k-2} commutes through every (T_j+1) for $j \geq k$ (index gap ≥ 2).

The case-by-case derivation matches May-15 Sec. 4 with no shape-dependent ingredient. $\square$

4 The same-row obstruction and the formula

By the May-19 Phase A theorem, $R'_n V_\lambda = E_+|_{V_\lambda} = \Phi_*(V_{\mu_1}) \oplus S$ (Lemma 3). Therefore:

$$\begin{aligned} B_{\ell+1}^{(\lambda)} V_\lambda &= R'_{\ell+1} \cdots R'_{n-1} (R'_n V_\lambda) \\ &= R'_{\ell+1} \cdots R'_{n-1} (\Phi_*(V_{\mu_1}) + S) \\ &= R'_{\ell+1} \cdots R'_{n-1} \Phi_*(V_{\mu_1}) + R'_{\ell+1} \cdots R'_{n-1} S. \end{aligned}$$

By Lemma 6, the first summand equals the desired Φ_* -image with outer factors. The same-row summand is the **obstruction**: if it vanishes, the reduction formula at the B -level holds; if not, we get a strict containment.

Conjecture 7 (Same-row dies). *Under (H1)–(H3),*

$$R'_{\ell+1} R'_{\ell+2} \cdots R'_{n-1} S = 0,$$

where S is the same-row part of $E_{n-1}^+|_{V_\lambda}$ (Lemma 3).

Theorem 8 (Conditional reduction formula). *Under (H1)–(H3), assuming Conjecture 7:*

$$B_{\ell+1}^{(\lambda)} V_\lambda = (T_\ell+1)(T_{\ell+1}+1) \cdots (T_{n-2}+1) \Phi_* \left(B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1} \right). \quad (3)$$

In particular,

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda \leq \dim B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1}. \quad (4)$$

Proof. Immediate from the decomposition above plus Lemma 6 and Conjecture 7. The dim bound uses injectivity of Φ_* and dimension-non-increase under linear maps. $\square$

4.1 Trivial case: shapes without same-row pairs

Corollary 9 (Reduction is unconditional when $S = 0$). *If λ admits no same-row pair $(i, \lambda_i - 1), (i, \lambda_i)$ with $\lambda_i \geq 2$ and $\lambda_{i+1} \leq \lambda_i - 2$ or $i = \ell$, then $S = 0$ and the reduction formula (3) holds unconditionally.*

This corollary applies to:

- Hooks $(k, 1^{n-k})$ with $k \geq 2$: $\lambda_1 = k, \lambda_2 = 1 \leq k - 2$ only if $k \geq 3$, otherwise $\lambda_2 = 1 > k - 2$. At $k = 3$: $\lambda_2 = 1 \leq 1 = k - 2$. Same-row pair $(1, 2), (1, 3)$ at row 1 is valid? We need $(2, 2)$ to not constrain — and indeed $\lambda_2 = 1$ so $(2, 2) \notin \lambda$. So at $k = 3$, same-row IS valid; not covered by this corollary.

Wait, this contradicts Remark 5 for $(2^a, 1^*)$ which IS covered. Let me re-check.

Remark 10 (Re-examination of which shapes are covered). The same-row pair at row i is valid iff $\lambda_i \geq 2$ AND $\lambda_{i+1} \leq \lambda_i - 2$ (so $(i+1, \lambda_i - 1) \notin \lambda$) OR $i = \ell$. For:

- $(2^a, 1^{n-2a})$: rows have $\lambda_i = 2$ for $i \leq a$, $\lambda_i = 1$ for $i > a$. At row $i \leq a$: need $\lambda_{i+1} \leq 0$, so $i = a$ and $\ell = a$ (i.e., no trailing 1s, $n = 2a$). Generic case has $n > 2a$, so $\ell > a$, no same-row. $S = 0$.
- $(k, 1^{n-k})$ at $k \geq 2$: row 1 has $\lambda_1 = k$, $\lambda_2 = 1$. Same-row at row 1 valid iff $1 \leq k - 2$, i.e., $k \geq 3$. So:

- Hook $(2, 1^{n-2})$: no same-row, $S = 0$.
- Hook $(k, 1^{n-k})$ for $k \geq 3$: same-row at row 1 is valid, $\dim S = f^{((1, 1^{n-2}))} = ?$. Hmm wait letting me think.

- $(3, 3, 1^q)$: row 2 has $\lambda_2 = 3$, $\lambda_3 = 1 \leq 1$. Same-row at row 2 is valid. $\dim S = f^{(3, 1^{q+1})}$.

So Corollary 9 covers $(2^a, 1^*)$ (giving the May-15 result back), but NOT $(3, 3, 1^*)$ nor hooks $(k, 1^*)$ for $k \geq 3$. The latter is significant: even hooks need Conjecture 7 for the May-15-style reduction.

5 Computational verification of the conjecture

5.1 The $(3, 3, 1^q)$ family

Proposition 11 (Same-row dies for $(3, 3, 1^q)$, computational). *For $\lambda = (3, 3, 1^q)$ at $q \in \{2, 3, 4, 5\}$ (so $n \in \{8, 9, 10, 11\}$),*

$$R'_{\ell+1} R'_{\ell+2} \cdots R'_{n-1} S = 0.$$

Proof (computational). Mod- P Gaussian elimination at $P = 100\,003$, $q = 11$. Scripts at `~/projects/scratch/2020`. The iteration ranks at $q = 4$ ($n = 10$, $\ell = 6$):

$$\dim S = 21 \xrightarrow{R'_9} 6 \xrightarrow{R'_8} 1 \xrightarrow{R'_7} 0.$$

Analogous decay verified at $q \in \{2, 3, 5\}$. □

Theorem 12 ($\dim B = 2$ for $(3, 3, 1^q)$, $q \in \{2, 3, 4, 5\}$). *For $\lambda = (3, 3, 1^q)$ at $q \in \{2, 3, 4, 5\}$,*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = 2.$$

Proof. Direct mod- P computation via `compute_dim_B.py` at the four values of q . Cross-verification: the dim equals $f^{\lambda^{(\geq 2)}} = f^{(2,2)} = 2$ at each q , agreeing with the May-13 closed-form prediction.

The structural piece: by Lemma 6 + the same-row proposition above, $B_{\ell+1}^{(\lambda)} V_\lambda = (T_\ell + 1) \cdots (T_{n-2} + 1) \Phi_*(B^{(\mu_1)})$. By May-12 evening at $q - 1 \geq 3$ (i.e., $q \geq 4$), $\dim B^{(\mu_1)} = 2$, giving the upper bound $\dim \leq 2$. The lower bound $\dim \geq 2$ comes from the same computation.

At $q \in \{2, 3\}$, May-12 evening does not apply directly (it requires $|\mu_1| \geq 8$, i.e., $q \geq 4$); the result is purely computational at these q . □

5.2 Status of the same-row-dies conjecture

What we know. Conjecture 7 is verified computationally for $(3, 3, 1^q)$ at $q \in \{2, 3, 4, 5\}$. The decay pattern $21 \rightarrow 6 \rightarrow 1 \rightarrow 0$ (at $q = 4$) suggests the same-row part lives inside an iterated E_1^- -leakage structure that drops by roughly a factor of 3 per step until terminating.

What we don't know. A structural proof. The natural attempt — identify $S \cong V_{(3,1^{q+1})}$ on $n-2$ letters via the SYT correspondence and apply the j -formula to the inner shape — is complicated by the factor $(T_{n-2}+1)$ in R'_{n-1} , which moves letter $n-1$ out of its position at $(2,2)$ and so takes us out of S between iterations. The image is no longer an $H_q(S_{n-2})$ -submodule of V_λ corresponding to $V_{(3,1^{q+1})}$ after one R'_{n-1} application.

Where the gap really lives. The same-row part S corresponds to the LR-summand $V_{(2,1^{q-1})} \boxtimes V_{(3,1^{q+1})}$ inside the parabolic restriction (or some related sub-structure) at the $\{c_1, c_*\}$ -pair level. Its decay under iterated R' is parallel to a sub-version of the rank-zero j -formula on $V_{(3,1^{q+1})}$ — but the precise correspondence is not a direct $H_q(S_{n-2})$ -iso because the iteration mixes the same-row part with letters $\leq n-2$ via T_{n-2} .

6 What was learned

1. **The May-15 reduction does NOT verbatim generalise to all $r = 1$ shapes.** The naive generalisation fails for shapes with same-row corner pairs, where $E_+ \supsetneq \Phi_*(V_{\mu_1})$. The $(3, 3, 1^q)$ family is a clean witness: E_+ has 21 extra dimensions at $q = 4$.
2. **The algebraic commutation on $\Phi_*(V_{\mu_1})$ does generalise.** Lemma 6 carries through unchanged.
3. **The full reduction formula salvages, conditionally.** If the same-row part dies under iterated R' (Conjecture 7), the formula holds. Verified computationally for $(3, 3, 1^q)$ at small q .
4. **$\dim B = 2$ for $(3, 3, 1^q)$ at $q \in \{2, 3, 4, 5\}$.** Theorem 12: structural ≤ 2 via the conjecture plus May-12 evening at $q \geq 4$, plus mod- P at $q \in \{2, 3, 4, 5\}$.
5. **The structural same-row-dies claim is the natural next target.** It's verified across multiple q but no proof yet. The j -formula approach to $(3, 1^{q+1})$ does not transfer directly. A correct framing might involve the $H_q(S_{n-2})$ -module structure of the FULL E_+ rather than just its Φ_* -piece.

7 Honest status

This session began with the PROVE.md target of proving the decoration-shape isomorphism $B \cong V_{\lambda(\geq 2)}$. That target was refuted earlier today (2026-05-13-decoration-iso-failed.tex, commit 391bb50).

The session's pivot was an attempt to extend the May-15 inductive reduction from $(2^a, 1^*)$ to all $r = 1$ shapes. The first draft of this paper claimed this generalisation verbatim. The claim turned out to be *false* as stated: $(3, 3, 1^q)$ has a same-row part in E_+ that breaks the May-15 setup. The honest content of this paper is:

- Isolation of the same-row obstruction (Section 4).
- A conditional reduction formula (Theorem 8), conditional on Conjecture 7.
- Computational confirmation that the conjecture holds for $(3, 3, 1^q)$ at small q .
- $\dim B = 2$ for the $(3, 3, 1^q)$ family at $q \in \{2, 3, 4, 5\}$ (Theorem 12).

The structural proof of Conjecture 7 remains the cleanest next target on this thread.